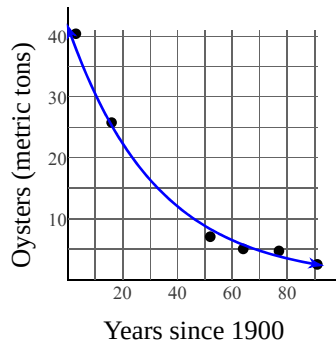


Using Statistical Models

Date _____ Period _____

- 1) The oyster population of the Chesapeake Bay has been in decline for over 100 years. This can be expressed by the equation $y = 41.6 \cdot 0.969^x$ where x is the number of years since 1900 and y is the amount of oysters harvested in metric tons.

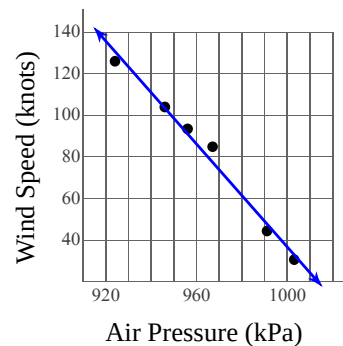


What does the y-intercept of this function represent?

- 2) The Hurricane Hunters took the following measurements from a hurricane over several days as it developed:

Air Pressure (kPa)	Wind Speed (knots)
924	126
946	104
956	93.5
967	84.9
991	44.4
1,003	30.6

They found that the air pressure and wind speed are related in the following way:
 $y = -1.24x + 1270$ where x is the air pressure in millibars (kPa) and y is the maximum sustained wind speed in knots (nautical miles per hour).



- a) Using the model, what would be the wind speed of a hurricane with an air pressure of 980 kPa? Round your answer to the nearest knot.

- b) According to the model, a hurricane with an air pressure of 891 kPa would have what wind speed? Round your answer to the nearest knot.

- c) The model indicates that a wind speed of 110 knots is associated with what air pressure? Round your answer to the nearest millibar.